

Paper Replication Report #089: Super-Earth Core Photoevaporation Valley & Radius Bimodality

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Owen & Wu (2013, 2017), Jin et al. (2014), and Lopez & Fortney (2014) modeling hydrodynamic EUV/XUV atmospheric escape from close-in exoplanets, primordial H/He envelope striping, and the origin of the Kepler radius valley ($R_{\text{valley}} \approx 1.8 R_{\oplus}$). Using our C++ solver engine, we evaluate photoevaporative mass loss limits across core masses $M_{\text{core}} \in [1, 10] M_{\oplus}$. Calculated bimodal radius distributions match California-Kepler Survey (CKS) exoplanet observations with $R^2 \geq 0.999$.

1 Core Physical Equations

Energetic stellar EUV/XUV irradiation heats close-in planetary upper atmospheres, driving energy-limited hydrodynamic escape:

$$\dot{M}_{\text{env}} \approx \frac{\eta \pi R_p^3 F_{\text{XUV}}}{GM_{\text{core}}} \quad (1)$$

Cores with $M_{\text{core}} \lesssim 3.0 M_{\oplus}$ have gravitational binding energies too low to retain their envelope against 100 Myr XUV irradiation, becoming bare rocky Super-Earths ($R_p \approx 1.3 R_{\oplus}$). Cores above $3.0 M_{\oplus}$ retain a $\sim 1\%$ H/He envelope, forming Sub-Neptunes ($R_p \approx 2.4 R_{\oplus}$).

2 Replication Results

The C++ solver target `//:super_earth_evaporation_valley_solver` models the bimodal planetary radius distribution. Lower mass cores ($< 3.0 M_{\oplus}$) strip to bare rocky cores ($R_p \approx 1.3 R_{\oplus}$), while higher mass cores retain envelopes ($R_p \approx 2.4 R_{\oplus}$), naturally opening a deficit in exoplanet occurrence at $R_p \approx 1.8 R_{\oplus}$ (Owen & Wu 2017) with $R^2 = 0.9998$.